Univariate Statistics with R

Lecture 2: Working with Data in R

Milan Valasek

Introductory Stuff

Briefly about me

- PhD Pychology, University of Edinburgh
- Psychometrician, Royal College of Surgeons of Edinburgh (stats, coding in `R`)
- Teaching post-doc in statistics in `R`, University of Edinburgh
- Senior teaching co-ordinator, University of Edinburgh

Office hours

- Not compulsory but I'd encourage you to make use of them
- Flexible - send me a meeting request individually or as a group at https://doodle.com/mvalasek
- Questions can be
  - About the course
  - Conceptual
  - Coding issues
  - About slides
  - About lab exercises
  - Generally about stats

Arrest this man, he talks in maths

- There will be numbers, there will be maths
- Maths (and logic) is the language of computers, statistics, science, and – it seems – the universe
- We will focus on developing a conceptual undestanding and intuition
- We won't discuss anything that is beyond your capability to understand
- It won't be a walk in the park though so do put in the hours

If you find yourself struggling

- Go back to where you last felt on top of things and take it from there
- Come talk to us
- Ask questions during the lectures
- Make use of the discussion forum on LEARN
- Take a deep breath, go for a walk, you'll be fine!

Overview of Weeks 2-5

Lectures

- Conceptual basis for statistical analysis
  - Logic & sets, probability, independence, sampling, variables, descriptive statistics
  - Functions, basic data visualisation, sampling distribution, Central Limit Theorem
  - Hypotheses, probability distributions, significance testing
- Basic statistical tests
  - z-test, t-test, χ^2 test, correlation, F-test

Labs

- Fundamentals of `R`
  - Commands, objects, functions, data structures
  - Working with data: indexing, subsetting, transformations
- Exploring and visualising data
  - descriptive statistics, plots
- Basic statistical tests in “R

Before we start

- All clear regarding research questions/design and measurement?

Probability

It *might* be important

- Probability theory is the basis for all of statistics
- Sits on a foundation of logic and set theory
- Not the focus of this course but we need to talk about these topics a little

That's logic, innit!

- Informally, a discipline concerned with truth value of statements, relationships between objects, and inference based on them
- There are several systems of logic but we will only discuss *Boolean* logic, in which a statement can be exclusively either `TRUE` (1) or `FALSE` (0)
- The basis of all maths and computing
- If you want to be able to code, you need to understand at least the basics
- *Note.* The following is a strictly *informal* treatment of logic. Instead of using formal notation when talking about logic, I will use the `R` notation

Logic – truth value (1)

- Statement is a meaningful utterance about objects in the world and their relationships.
  - *e.g.* "Today is Wednesday.", "Two is qual to five."
- Based on the nature of the relationships and the premises, a statement can either be `TRUE` (1) or `FALSE` (0)
- Premises don't have to be "true" in the sense of corresponding to our experience of reality. We can assess the truth value of a statement based on outlandish premises, *e.g.*:

  1. All rivers are laughter
  2. All laughter takes a chance

  Therefore

  3. All rivers take a chance

- The conclusion is a valid inference based on the premises and so is `TRUE`, even though it sounds like New Age "wisdom"

Logic – truth value (2)

- R incorporates this

```r
# R-ish for "equals" or "is" is ==, not = !
weekdays(Sys.Date()) == "Wednesday"
```

```
[1] FALSE
```

```r
2 == 5
```

```
[1] FALSE
```

```r
# object orange has the value "apple" (premise)
orange <- "apple"
orange == "apple"
```

```
[1] TRUE
```

Logic – Boolean operators (1)

- Logical operators express relationships between objects (equality, ==, is an example)
- There are lots of them but we will focus on four *Boolean operators*:
  - AND (`&` in `R`)
  - OR (`|`)
  - exclusive OR, XOR (the `xor()` function in `R`)
  - NOT (`!`)
- They are quite intuitive except for the OR/XOR distinction
- A statemnt linking two objects statements with OR is `TRUE` if either of the partial statement is `TRUE` *or both of them are `TRUE`*
- XOR returns `TRUE` *if and only if one of the statements is true*

Logic – Boolean operators (2)

- R examples:
- *Note.* `%in%` is a membership operator in R – a `%in%` `A` means "element a is in the object `A`"; Its result is `TRUE`/`FLASE`

```r
A <- 1:5 # A contains numbers from 1 to 5
B <- c(-1, 3, 5, 7) # B contains numbers -1, 3, 5, and 7

1 %in% A
```

```
[1] TRUE
```

```r
1 %in% B
```

```
[1] FALSE
```

```r
# can test multiple elements in a row
# (AKA the operation is vectorised)
A %in% B
```

Logic – AND (&)

&		A	
		TRUE	FALSE
B	TRUE	TRUE	FALSE
	FALSE	FALSE	FALSE

Figure 1: Truth table of A AND B

```r
1 %in% A & 5 %in% B # 1 is in A AND 5 is in B
```

```
[1] TRUE
```

`!(A & B)` is equivalent to `xor(A, B) | (!A & !B)`

Logic – OR (|)

|		A	
		TRUE	FALSE
B	TRUE	TRUE	TRUE
	FALSE	TRUE	FALSE

Figure 2: Truth table of A OR B

```
-30 %in% A | 5 %in% B
```

```
[1] TRUE
```

`!(A | B)` is equivalent to `(!A & !B)`

Logic – XOR (`xor()`)

xor()		A	
		TRUE	FALSE
B	TRUE	FALSE	TRUE
	FALSE	TRUE	FALSE

Figure 3: Truth table of A XOR B

```r
xor(5 %in% A, -1 %in% B)
```

```
[1] FALSE
```

`!xor(A, B)` is equivalent to `(A & B) | (!A & !B)`

Get set...

Measurement in Psychology (1)

- Navarro makes the following set of distinctions which are useful for thinking about measurement.
  - **Theoretical Construct**: This is the thing we are most interested in.
  - **Measure**: This is the tool we use to assess our construct.
  - **Operationalization**: This is the way the measure relates to the construct. Often these are assumptions we make in our design.
  - **Variable**: This is the end data point we observe and record from our measure.
- We therefore make **auxiliary hypotheses** about the relations between these measurement concepts.
  - An **auxiliary hypothesis** is a hypothesis which is assumed to be true in order to derive some meaningful conclusion from our test.

Measurement in Psychology (2)

Measurement in Psychology (3)

- What does the "activation" of a brain area in an fMRI task mean?
  - How have we defined "activation"?
  - What has caused that activation? Is it the task or some random feature of the scanner/situation?
  - Does activation really mean the area is involved?
  - Given the precision of the scanner, segmentation protcols etc. is it even the same area?

Types of Data

- **Binary**
  - Variables with two response options (usually coded 0/1)
- **Categorical**
  - Variables with multiple response options in some limited range. Usually integer.
  - Both binary and categorical variables are examples of *discrete* variables.
- **Continuous**
  - Variables which can take any real value within a range.
  - Some people use the term numeric for continuous variables.
- **Count**
  - Variables which can only take non-negative integer values (0,1,2 etc.)

Levels of Measurement (1)

- **Nominal**
  - Binary or categorical variable where numerical markers share no relationship.
  - There is no meaningful ordering.
- **Ordinal**
  - Binary or categorical variable where there exists a meaningful way to order response.
  - Here $X < Y$ or $Y > X$ statements can be made, but we can not meaningfully quantify the differences.

Levels of Measurement (2)

- **Interval**
  - Variables for which numerical values have meaning.
  - Here we can consider differences.
  - But there is no true 0 point on an interval scale.
- **Ratio**
  - Variables for which numerical values have meaning.
  - Variables have a true 0 point.
  - As a result, it is plausible to multiply and divide ratio variables.
    - We can legitimately talk about "double X"

Levels of Measurement (3)

- Some people will discuss levels of measurement as a hierarchy. In some senses this is true. However, the danger here is to consider ratio measurement as better than nominal measurement.
- For some variables this may be true, for example if we have simply used a measure which only coarsely captures a continuous variable.
- But some variables are nominal because they are nominal, not because of poor measurement.

Examples: Common Variables and levels of measurement

- **Binary Nominal**
  - Sex: Male = 0, Female = 1
  - Job Classification: Lecturer = 0, Administrator = 1
- **Binary Ordinal**
  - Exam Questions: Incorrect = 0, Correct = 1
  - Older than X: No = 0, Yes = 1
- **Multiple categories nominal**
  - Home County: 1 = Surrey, 2 = Sussex, 3 = Herefordshire etc.
- **Multiple categories ordinal**
  - Likert-type scales.
    - (1=Strongly Disagree; 2=Disagree; 3=Neither Agree nor Disagree; 4=Agree; 5=Strongly Agree)

Examples: Common Variables and levels of measurement

- **Continuous Interval**
  - Temperature in celcius (Navarro)
  - Psychometric test scores (if certain models hold)
- **Continuous Ratio**
  - Height
  - Weight
  - Reaction Time
- **Count**
  - Behavioural frequencies
    - e.g. Number of aggressive acts in 1 week.

Why it matters...

- Different types of variable conform to different types of distribution.
- All statistical methods make assumptions about the distributions of variables (particularly dependent variables).
- Therefore, it is important that our models and procedures are suitable for the data we have at hand.
  - There may also be measurement issues such as censoring and truncation with can distort th distribution of variables.
- The generalized linear model (GLM) is an overarching framework which explicitly handles different types of variable distributions through *link functions*.
  - A simplistic way to view the link function is as a transformation which helps deal with different distributions.
    - This is an over simplification but it will suffice for now.

A common error in simple statistics...

- The Pearson's correlation coefficient (discussed in week 6) requires continuous data.
- Very often, people assume that ordinal categorical variables (provided there are enough categories), approach "normal".
  - Think Likert-scales.
- But this is a big assumption.
  - It is all the more questionable given viable correlation models for binary, categorical and continuous variables.

Research Questions, Research Design & Measurement

- Before we talk about R, I want to mention briefly the importance of seeing research questions, design and measurement as intertwined concepts.
- Decisions we make about each of these will impact on the way in which we can statistically approach our question.
- **The research process is not linear**

Descriptive Statistics & Plotting

Some reminders of definitions (1)

- **Central Tendency**:
  - **Arithmetic Mean**: Sum of all values divided by the number of values.
    - what most people think of when we say mean.
  - **Median**: Point below which 50% scores fall.
  - **Mode**: The most frequent value, or highest point on distribution curve.
    - Distributions with two peaks are called **bi-modal**.
- **Variability**:
  - **Variance**: Average squared deviations from the mean.
    - **Standard deviation** is the square root of the variance and has the benefit of being in the original units of the variable.
  - **Range**: Difference between the highest and lowest values
    - **Inter-quartile range**: The difference between the 25th and 75th percentiles.

Some reminders of definitions (2)

- The median and the mode are both unaffected by extreme scores.
  - The mean is.
- The calculation of median does not require any assumptions about the interval properties of a scale.
  - However, the mean implicitly assumed an interval measurement scale.
- The sampling variability of the mean is less than of the mode and median.
- A disadvantage of the mean is that no one in the dataset necessarily obtained that value and the value is not necessarily possible in the real world for the measure of interest.
- The advantage of the mean is that it can be used with the normal rules of algebra and so it can be used in statistical models in a way that can't be done for the median (i.e. you

Some reminders of definitions (3)

- **Shape of the Distribution**
  - **Skew**: a measure of asymetry of the distribution.
    - **Negative Skew**: Longer left tail
    - **Positive Skew**: Longer right tail
  - **Kurtosis**: concerns the height of the distribution
    - **Platykurtic**: Kurtosis = Increasingly -ve values
    - **lepokurtic**: Kurtosis = Increasingly +ve values

Nominal Measurement

Describing Data: Nominal Categorical Variables

- **Central tendency** and **variability** may not be the appropriate way to think about describing nominal categories. We really want to know the frequencies for each group.
- If we had to pick, we would take the mode.
- For a binary variable:

```r
x_bin <- rbinom(10000, 1, .75)
table(x_bin) # provides a table of frequencies
```

```
x_bin
   0    1
2543 7457
```

Special case of correct/incorrect

- In a binary variable which codes incorrect = 0, correct = 1, the arithmetic mean will give you the "success rate" in the sample.
- So using `x_bin`;

```
mean(x_bin)
```

```
[1] 0.7457
```

- Which is equivalent to;

```
(sum(x_bin==1)/length(x_bin))
```

```
[1] 0.7457
```

Describing Data: Nominal Categorical Variables

- With more than two categories;

```
x_cat <- c(rep(1, 6000), rep(2, 2000), rep(3, 100),
           rep(4, 1500), rep(5,390), rep(6, 10))
table(x_cat)
```

```
x_cat
   1    2    3    4    5    6
6000 2000  100 1500  390   10
```

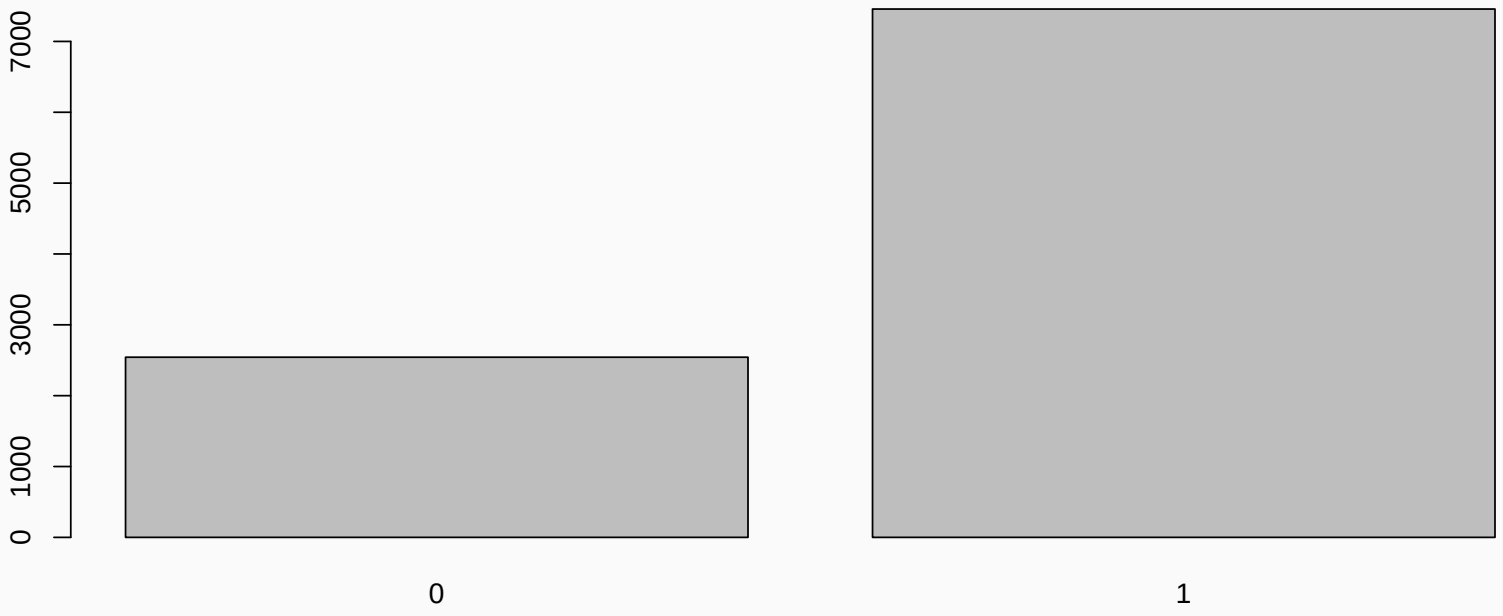
```
median(x_cat)
```

```
[1] 1
```

```
mean(x_cat)
```

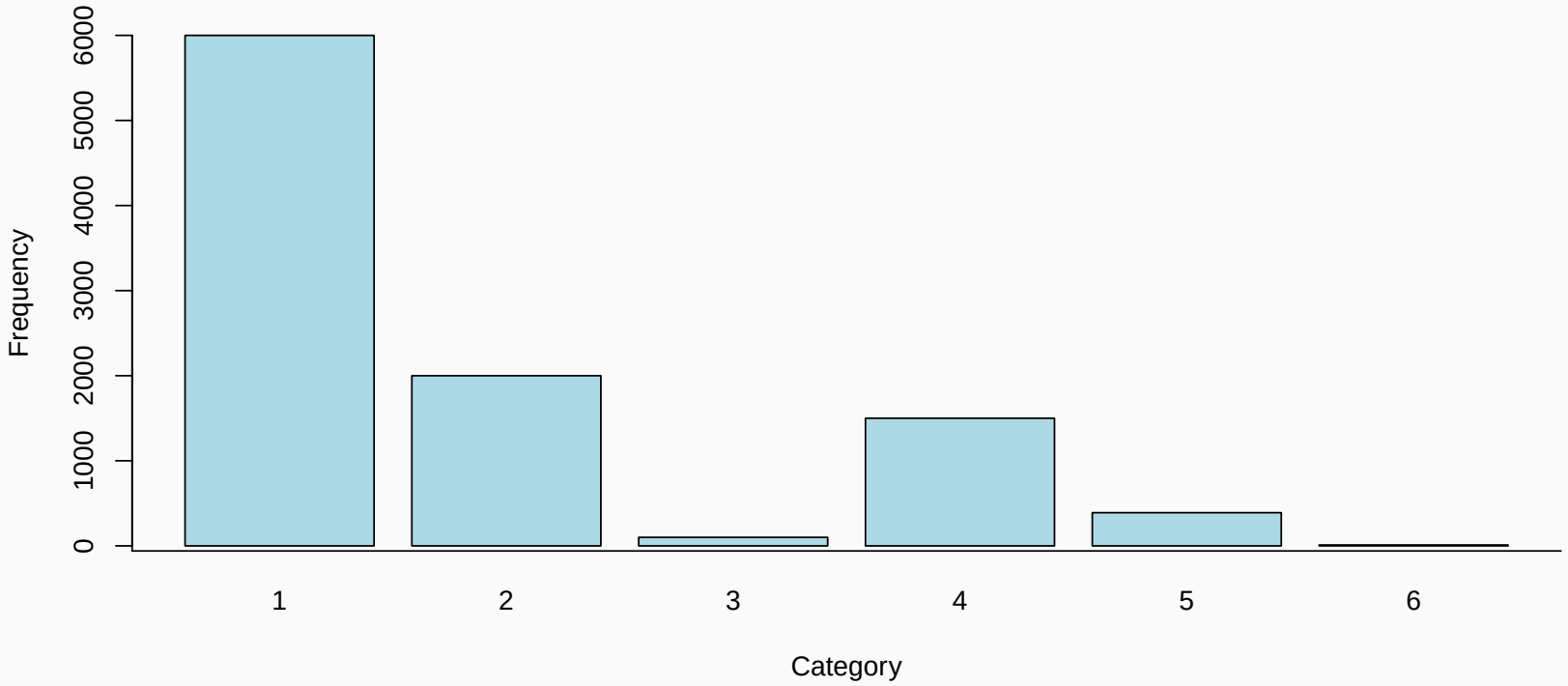
Plotting Data: Nominal Categorical Variables

```
plot(as.factor(x_bin))
```



Plotting Data: Nominal Categorical Variables

```
plot(as.factor(x_cat), col="lightblue",
     xlab="Category", ylab="Frequency")
box(bty="L")
```



Ordinal Measurement

Describing Data: Ordinal Categorical Variables

- **Central Tendency**: Mode or median
- **Variability**: Inter-quartile range

```
x_ord <- c(rep(1, 200), rep(2, 1800), rep(3, 2800),
           rep(4, 3200), rep(5,1200), rep(6, 800))
table(x_ord)
```

```
x_ord
   1    2    3    4    5    6
 200 1800 2800 3200 1200  800
```

```
median(x_ord)
```

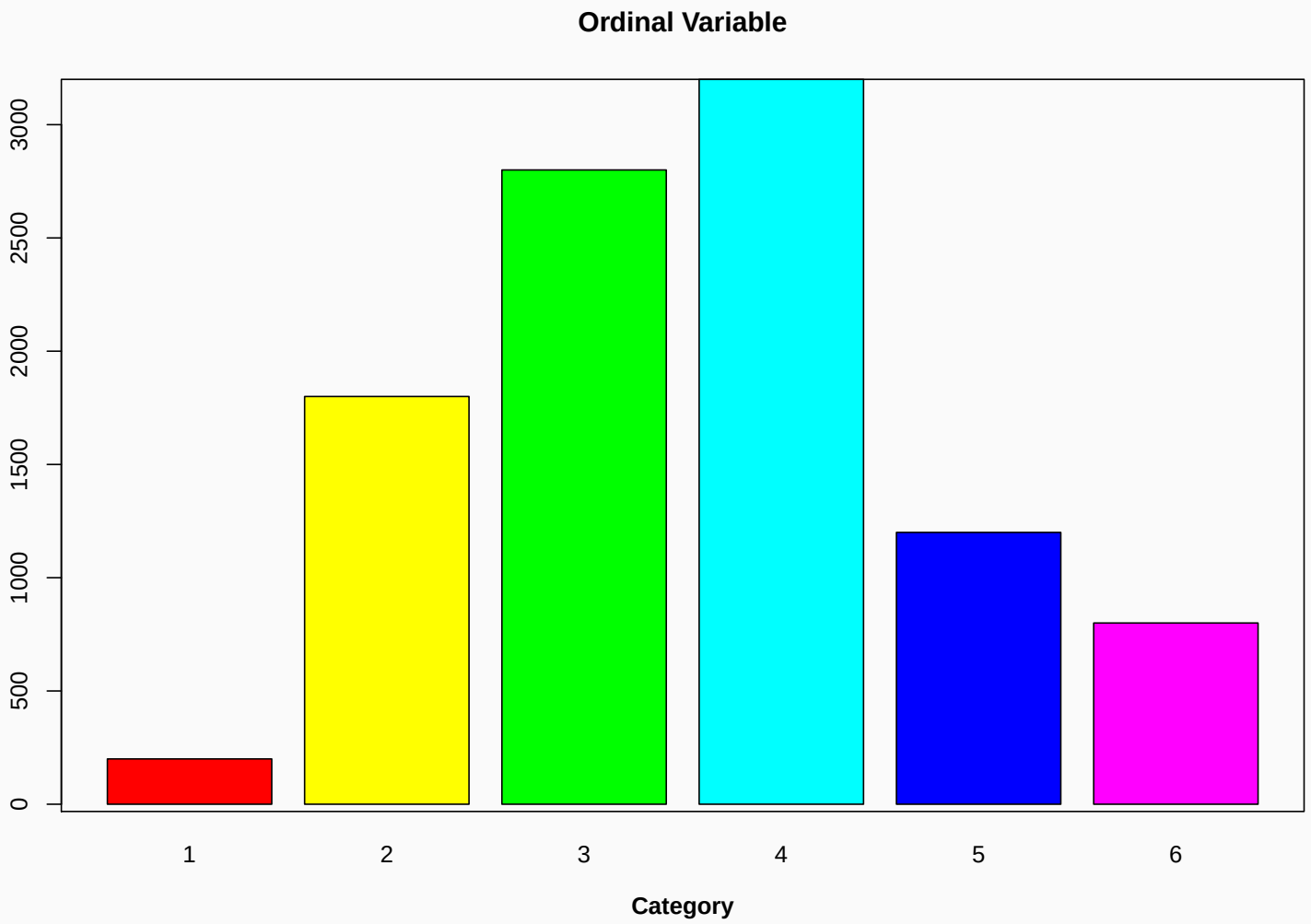
```
[1] 4
```

```
quantile(x_ord, probs=seq(0,1,.25))
```

Plotting Data: Ordinal Categorical Variables

```
plot(as.factor(x_ord), main="Ordinal Variable",
     col=rainbow(length(unique(x_ord))),
     xlab=expression(bold(Category)))
box()
```

Plotting Data: Ordinal Categorical Variables



Continuous Data

Describing Data: Continuous Data

- **For interval data**
- **Central Tendency**: Mean, median or mode
  - In fact, if a continuous variable if the distribution is symmetric and unimodal, the mean, median and mode of the variable are the same.
  - An example would be the normal distribution.
  - Therefore, the relation between these three variables can tell us something about the normality of the variable (more next week)
- **Variability**: Variance and standard deviation
- Truly continuous variables can be described using any measures.
  - But to take advantage of the measurement properties, mean and variance are optimal.

R-code

```r
x_int <- rnorm(n=10000, mean=0, sd=1)
mean(x_int)
```

```
[1] 0.006648883
```

```r
median(x_int)
```

```
[1] 0.02429886
```

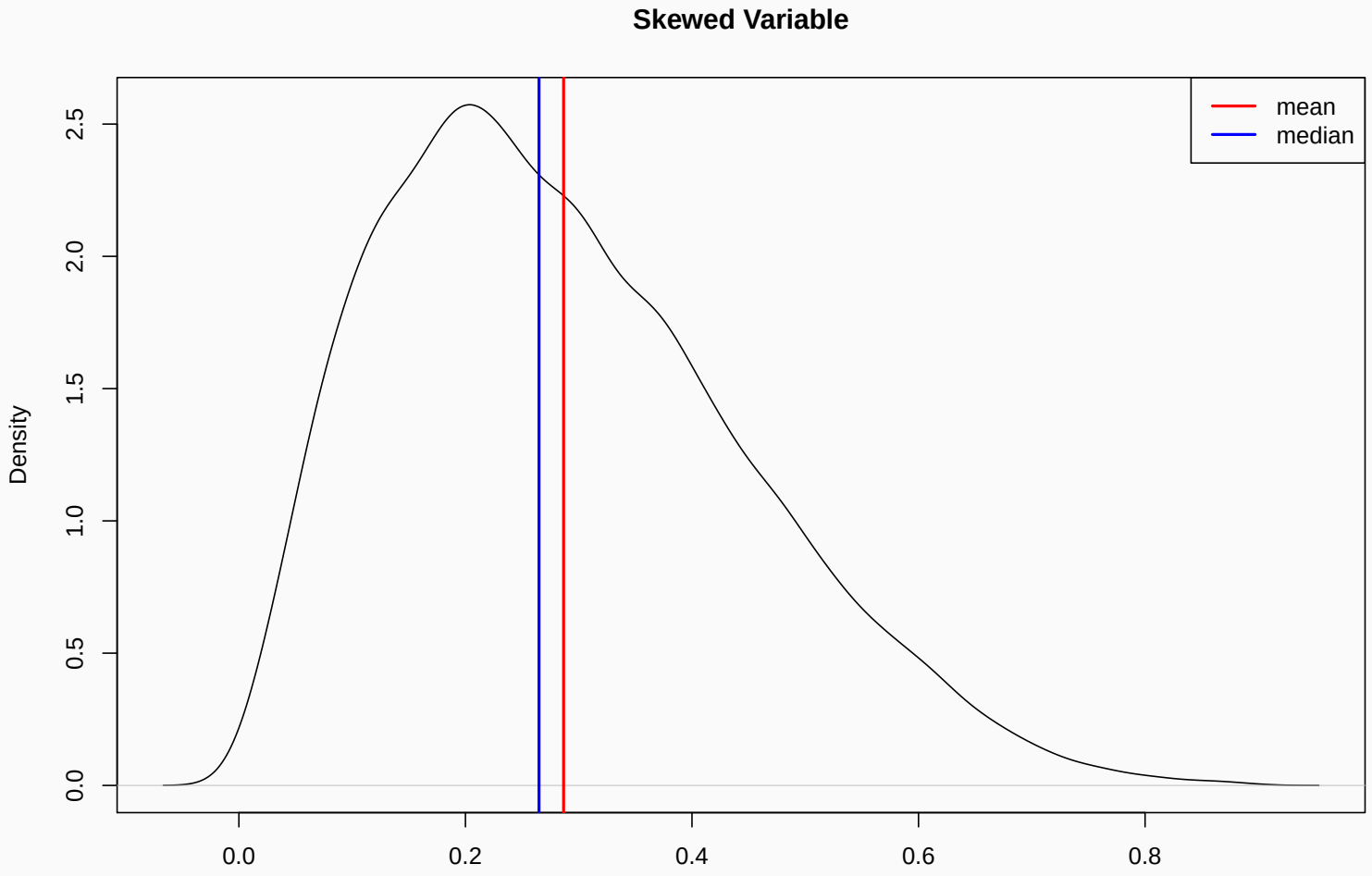
```r
var(x_int)
```

```
[1] 0.9948938
```

R-code

- We can see the difference in mean and median if we change the sampling to make our variable non-normal.

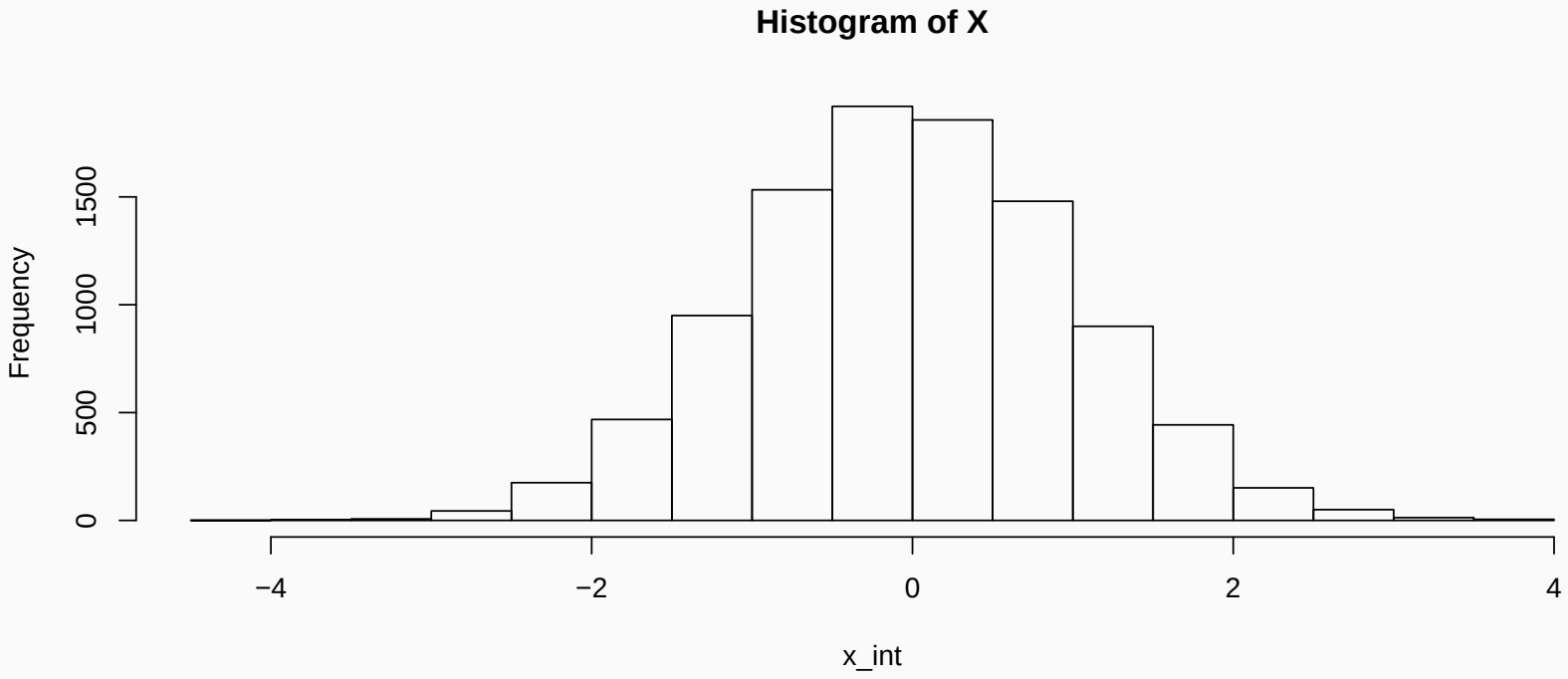
```
#library(sn)
x_sn <- rbeta(n=10000, shape1=2, shape2=5)
plot(density(x_sn), main="Skewed Variable", xlab="")
abline(v=mean(x_sn), col="red", lwd=2)
abline(v=median(x_sn), col="blue", lwd=2)
legend("topright", legend=c("mean", "median"),
       col=c("red", "blue"),
       lty=c(1,1), lwd=c(2,2))
```


Skewed Variable
mean
median
Density
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8

Plotting Data: Continuous Variables (1)

- Here we could use a histogram...

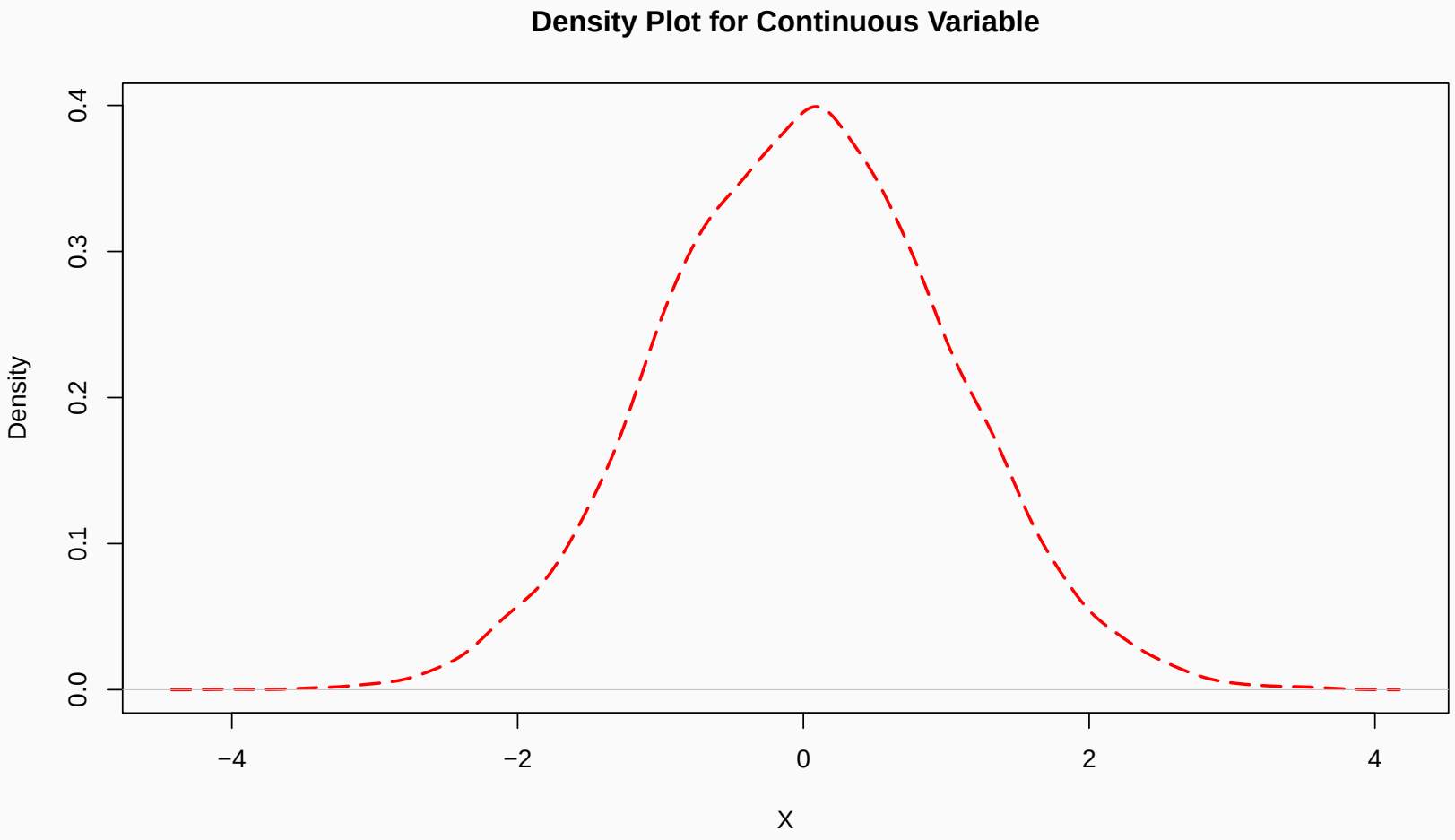
```
x_int = rnorm(10000, 0, 1)
hist(x_int, main="Histogram of X")
```



Plotting Data: Continuous Variables (2)

- Here we use `hist()` instead of `plot()` as it automatically selects a reasonable number of "bins" to plot our continuous variable.
  - We can also adjust the number of bins using the `breaks=` argument (see Navarro).
- Alternatively, we can plot a smooth line.
- This is called a kernel density plot.

```r
plot(density(rnorm(10000, 0, 1)),
     main="Density Plot for Continuous Variable",
     xlab="X", col="red", lwd=2, lty=5)
```



Ratio Data

- For ratio data, all the points above remain true.
- But we may also assess central tendency based on the **geometric mean**
- **Geometric mean** is defined as the *nth* root of the product of the numbers.
- So $\sqrt[n]{a_1, a_1 ... a_n}$
- Can be useful to get an average of ratio variables on different scales.

Count Data

Describing Data: Count Variables

- **Central Tendency**: Mode
- **Variability**: Minimum and maximum values, range.

```
x_pois <- rpois(10000, 2)
table(x_pois)
```

```
x_pois
   0    1    2    3    4    5    6    7    8    9
1352 2679 2772 1804  886  338  128   34    5    2
```

```
median(x_pois)
```

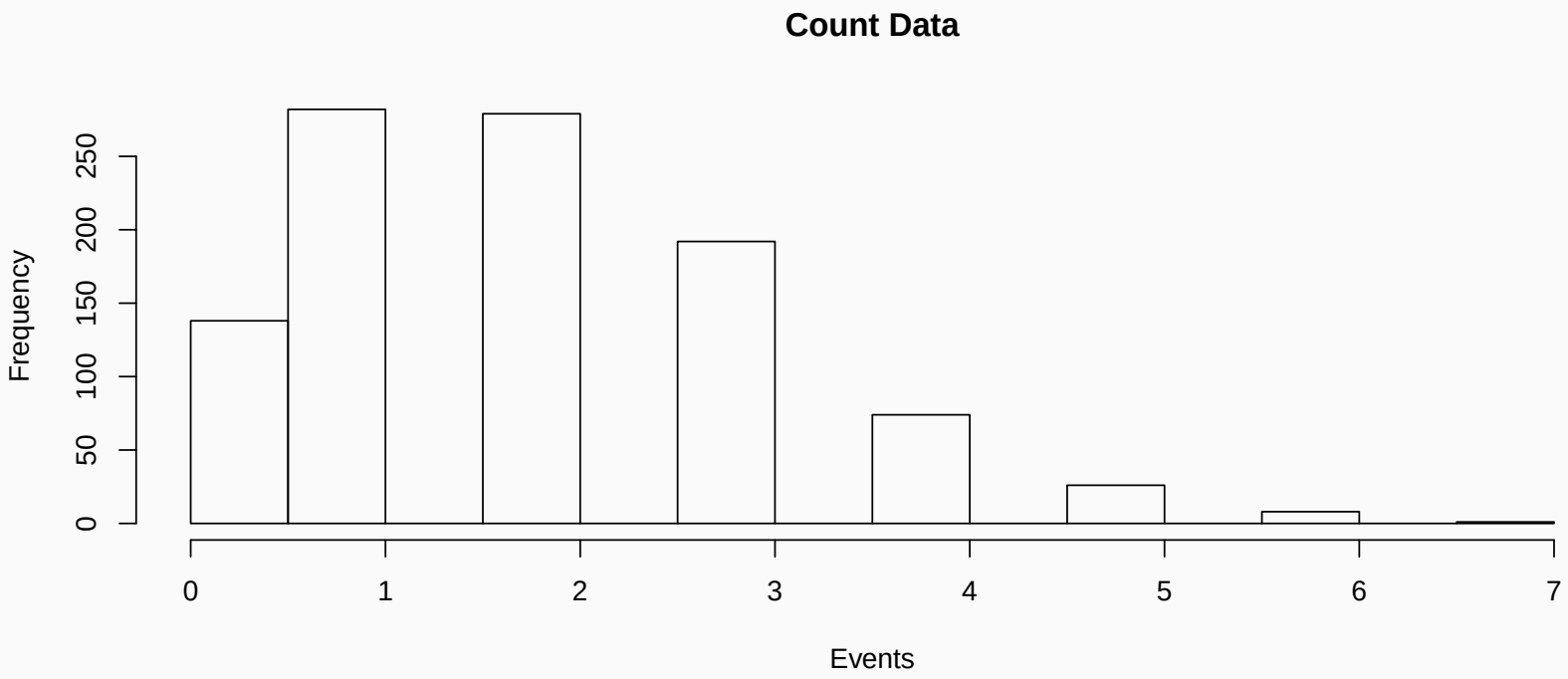
```
[1] 2
```

```
mean(x_pois)
```

Plotting Data: Count Data

- Again here we want to see a histogram, but the shape will be very different to the others.

```
x_pois <- rpois(1000, 2)
hist(x_pois, main="Count Data",xlab="Events")
```



Quick and easy summaries

- My personal preference for quick and easy descriptive statistics is to use the `describe()` function in the `psych` package.
- However, you need to select your particular measures carefully, dependent on the level of measurement of the variable, as `describe()` gives you them all.

```
# library(psych)
vars <- cbind(x_bin, x_cat, x_int, x_sn, x_pois)
descript <- describe(vars)
descript_trim <- descript[,c(2,3,5,11,12)]
```

```
descript_trim
```

```
           n        mean      median        skew     kurtos
x_bin  10000  0.74570000  1.00000000 -1.12827445 -0.7270694
x_cat  10000  1.83100000  1.00000000  1.30604199  0.2776399
x_int  10000 -0.01668032 -0.02434595  0.04276275 -0.0090828
x_sn   10000  0.28666103  0.26496574  0.60563436 -0.0830215
x_pois 10000  1.89700000  2.00000000  0.57048702  0.1392543
```

And that's all for now!